

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0670

Fiesta 138 kV
August 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0670. The Interconnection Customer proposes a 77.0 MW solar photovoltaic facility interconnecting at the Fiesta 138 kV substation in the Virginia area. The request was received on August 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 226 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,932,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SKAGGS ISLAND JUNCTION 2 to MITCHELL DAM 345.0 kV, the FIESTA to LANEIA 138.0 kV loads to 113.5% of its 175 MVA rating. Project DFAX: 60.5%.

The TIPPECANOE to SKAGGS ISLAND JCT 2 138.0 kV Line is overloaded to 107.6% of its 238 MVA emergency rating for the loss of the SKAGGS ISLAND JCT 2 to CHANCELLOR ROW 345.0 kV line. The project contributes 11.3% DFAX (8.7 MW) to this overload.

TIPPECANOE to OLD CHUCH 138.0 kV: 102.6% loading (226 MVA) following the outage of the CHANCELLOR ROW to SKAGGS ISLAND JCT 2 345.0 kV; DFAX = 11.3%.

The MORRELL to OLD CHURCH 138.0 kV is overloaded to 102.2% of its 197 MVA emergency rating for the loss of the CHANCELLOR ROW to SKAGGS ISLAND JUNCTION 2 345.0 kV. The project contributes 11.3% DFAX (8.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FIESTA to LANEXA 138.0 kV	Rebuild line with higher-capacity conductor	\$5,702,000	53% (\$3,030,000)
TIPPECANOE to SKAGGS ISLAND JCT 2 138.0 kV Line	Reconductor line	\$1,762,000	12% (\$210,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TIPPECANOE to OLD CHUCH 138.0 kV	Reconductor line	\$2,932,000	12% (\$350,000)
MORRELL to OLD CHURCH 138.0 kV	Reconductor line	\$11,245,000	12% (\$1,342,000)
	Total allocated to project		\$4,932,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.